

Ashutosh Joshi

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PROFESSIONAL EXPERIENCE

Software Engineer - AI/ML Systems | NASA (via. SciGlob Instruments)

November 2024 – Present

- Built a full-stack instrument-health web application for NASA using Python/FastAPI, React/Next.js, and TypeScript, giving operators visibility into the health of 300+ instruments across five continents.
- Architected and deployed a production LLM inference platform on vLLM powering RAG and agentic workflows. Tuned continuous batching, chunked prefill, and prefix caching to raise throughput per GPU **87%** and cut P99 TTFT **20%**.
- Developed Go-based Kubernetes controllers and CRDs for model lifecycle management, GPU-aware scheduling, and autoscaling of vLLM inference workloads.
- Evaluated vLLM, SGLang, and TensorRT-LLM head-to-head on throughput, tail latency, and quantization support, and recommended the production runtime based on workload-specific benchmarks.
- Cut serving cost **60%** by shipping quantized execution (FP8/INT8, AWQ) and speculative decoding, evaluated against accuracy and latency targets before rollout.
- Built a benchmarking harness tracking TTFT, ITL, P95/P99 latency, and MFU/MBU, gating every model and config change on performance regressions.
- Scaled large-model serving across multi-GPU nodes with tensor and pipeline parallelism over NCCL. Deployed on Kubernetes with MIG partitioning for smaller models, cache-aware routing, and SLO-driven autoscaling, cutting idle GPU hours **40%**.
- Led root-cause analysis of latency regressions across multi-tier RAG and agent pipelines using Nsight Systems/Compute and PyTorch Profiler, resolving production incidents end-to-end.
- Leading a ground-up C++17 rewrite of the Blick spectral processing suite (L0→L2): **14** modular libraries, a parity test harness against legacy output, and a CUDA-ready GPU acceleration layer.
- Deployed deep-learning anomaly detection and an encoder-decoder CNN for real-time cloud detection across **300+** Pandora spectrometers in the Pandonia Global Network, eliminating hours of daily manual monitoring.
- Built edge-to-cloud ML pipelines and a full-stack fleet-health platform (FastAPI, React/Next.js, TypeScript) that parses raw L0 streams at the edge, scores instrument health 0–100, and delivers auditable trace-gas datasets to NASA/ESA researchers.

Data Analyst & Developer | 407 Associates LLC

April 2024 – November 2024

- Built Python data and ML pipelines (PySpark, scikit-learn, TensorFlow) and optimized SQL, AWS S3, and Snowflake workloads, cutting operating costs **25%**.
- Automated multi-source data cleaning and validation, and delivered internal web tools and Tableau/Power BI dashboards that replaced manual spreadsheet workflows.

System Engineer | Tata Consultancy Services Limited

June 2020 – August 2022

- Engineered ETL pipelines for Albertsons Co. processing **10M+** rows across multiple sources, with **20+** automated data-quality checks that cut data defects **30%**.
- Profiled and tuned cloud application performance under load, raising throughput **25%** and cutting response time **30%** while sustaining **99.5%+** availability.
- Built CI/CD deployment and monitoring automation across multiple environments, reducing release time **40%** and MTTR **25%**.

SKILLS

Languages: Python, C++17, GO Lang, CUDA, SQL, TypeScript, Bash

LLM Inference: vLLM, SGLang, TensorRT-LLM, Triton, KV cache, PagedAttention, continuous batching, prefix caching, chunked prefill, quantization (FP8/INT8, AWQ, GPTQ), speculative decoding, MoE serving

Performance & Profiling: TTFT, ITL/TPOT, P99 tail latency, MFU/MBU, benchmarking, Nsight Systems/Compute, PyTorch Profiler, root-cause analysis

Distributed GPU Systems: tensor & pipeline parallelism, NCCL, disaggregated prefill/decode, KV-cache offload, RDMA, GPUDirect, NVLink/InfiniBand

Kubernetes & Orchestration: CRDs, Helm, GPU scheduling, MIG, multi-tenant isolation, cache-aware routing, autoscaling

ML & GenAI: PyTorch, TensorFlow, scikit-learn, computer vision, time-series anomaly detection, RAG, pgvector, agentic AI, guardrails

Data & Backend: FastAPI, Flask, React/Next.js, PySpark, Snowflake, PostgreSQL, Redis, ETL/ELT pipelines

Cloud & DevOps: AWS, GCP, Azure, Docker, CI/CD, Prometheus, Grafana, OpenTelemetry, Linux

EDUCATION

- **M.S in Data Science (Advanced Applied AI)** | University of Maryland, Baltimore County
- **B.S in Computer Science Specialization in A.I.** | Medi-Caps University